

M002 Extension 4 — Complete the Maze

Topic: Add Two AND Conditions · **Course:** Maze Madness · **Level:** Extension (Intermediate) · **Time:** about 45 minutes

The agent knows **four rules** to follow a redstone trail. But the solver is **not finished**. At two junctions two signals are on at once, so one signal is not enough and the agent stops. You find the two signals and add two **AND** rules of your own. Then drive it to the goal.

Chat: l turn left, r turn right, run start solver, rl bring agent back.

1 · The Four Rules

Each rule checks **one** signal and does **one** action.

```
if agent.detect(REDSTONE, LEFT):
    agent.turn(RIGHT_TURN)
elif agent.detect(REDSTONE, RIGHT):
    agent.turn(LEFT_TURN)
elif agent.detect(REDSTONE, DOWN):
    agent.move(UP, 1)
elif agent.detect(REDSTONE, UP):
    agent.move(DOWN, 1)
```

Fill the table from the code. You will reuse these actions.

Signal	Action
redstone left	
redstone right	
redstone below (down)	
redstone above (up)	

2 · Find the Two Stuck Junctions

Open the **M002 Complete the Maze** world. Type `run`. Watch where the agent stops. Walk over and look around it.

Tick the signals on at each stuck junction.

	left	right	below	above
Junction 1				
Junction 2				

Only the first matching `elif` fires. Why does that leave the second signal unused?

3 · AND means BOTH

An **AND** is true **only when both** halves are true.

```
if agent.detect(REDSTONE, LEFT) and agent.detect(REDSTONE, DOWN):  
    ...
```

Example: left on but NOT below. Is the whole condition true?

Look at Junction 1 in your table. Write the AND condition that is true only there.

4 · Add the Two New Rules

Find the four rules in the code. Add **two** **elif** **branches**, one per stuck junction, using the signals **you** found. Each branch does the action for **each** signal.

Junction 1 — fill the blanks:

```
elif agent.detect(REDSTONE, _____) and agent.detect(REDSTONE, _____):  
    agent._____(_____)  
    agent._____(_____)
```


Junction 2 — write the whole branch yourself:

Put both new rules *before* the four single-signal rules. Why?

5 · Spot the Bug 🐛

This branch should fire **only when both** signals are on. Right now it fires when **either** one is on, so it triggers at wrong spots.

```
elif agent.detect(REDSTONE, LEFT) or agent.detect(REDSTONE, DOWN):  
    agent.turn(RIGHT_TURN)  
    agent.move(UP, 1)
```

Write the fixed code:

Why was it wrong?

6 · Finish the Whole Maze 📷

Type `run`. The goal is start to the **very end**, past both stuck junctions. Fix your rules and run again until it reaches the goal.

Send a photo or video of the agent at the goal. Then write 3 or 4 sentences with these starters.

Each of the four rules checked ...

The two signals at the first stuck junction were ... and ...

My AND rule was true only when both were ...

I put my new rules first because ...

7 · Plan Your Walkthrough Video 🎥

Film on your phone (or a parent's) while the agent solves the maze. Talk like you are teaching a friend. Try these words: **detect**, **redstone**, **AND**, **both**, **turn**, **move up**.

1. Show the maze and the redstone.
2. Type **run** and show the agent following it.
3. Read one AND rule and say the two signals it needs.
4. Show the agent reaching the goal.

Write what you will say. You can read it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.